

AI Worker Burnout – Linear Regression Analysis

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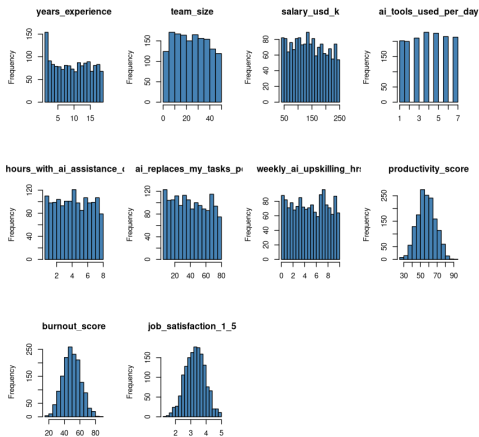
Dataset Overview



- 1500 employees, 21 variables
- No missing values
- Target: `burnout_score`

Key variables:

- `burnout_score` (16–86)
- `job_satisfaction` (1.3–5.0)
- `attrition_risk`



Correlations between Variables



Correlation Matrix of Numeric Variables



Research Question and Hypothesis



Can we predict burnout score from AI usage and job satisfaction?

Chosen predictors (from correlation analysis)

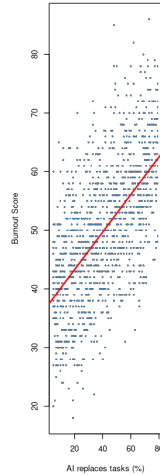
- `ai_replaces_my_tasks_pct` – positive effect ($r = 0.613$)
- `job_satisfaction_1_5` – negative effect ($r = -0.629$)
- `hours_with_ai_assistance_daily` – positive effect ($r = 0.386$)

Split: 90% training (1350 rows) / 10% test (150 rows)

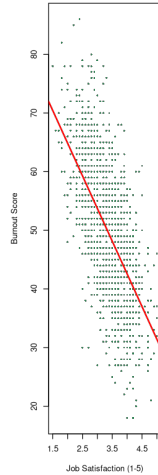
Predictors vs Burnout Score



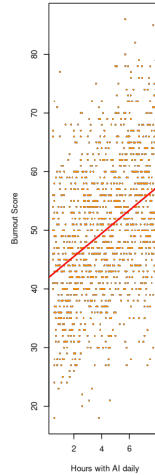
AI Task Replacement vs Burnout



Job Satisfaction vs Burnout



AI Hours vs Burnout



Linear Regression Model



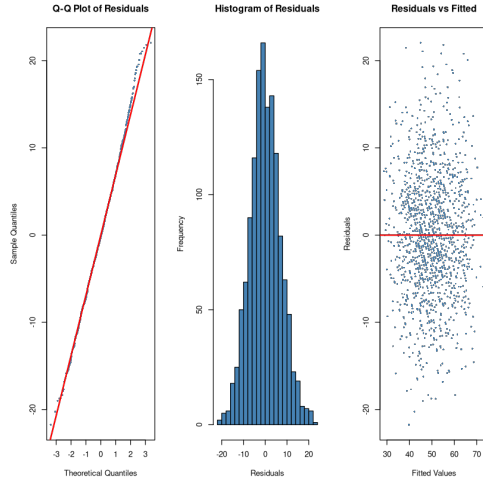
$$\hat{y} = 56.75 + 0.236 \cdot x_{\text{replace}} - 6.718 \cdot x_{\text{satisfaction}} + 1.418 \cdot x_{\text{hours}}$$

Metric	Value
R^2	0.611
Adj. R^2	0.610
Residual SE	7.10
F-statistic	705.2
p-value	$< 2e-16$

All three predictors significant at $p < 2e-16$

- Model explains 61% of variance in burnout
- Strongest predictor: job satisfaction

Assumption Checks



Residuals are approximately normal with no systematic pattern.

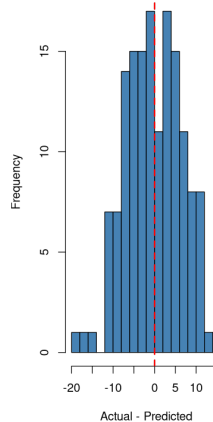
Test Set Evaluation



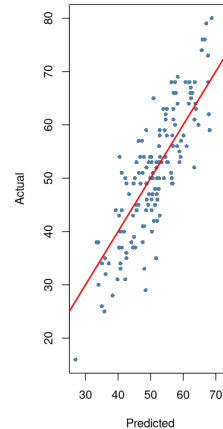
Statistic	Value
Mean	-0.18
Std dev	6.57
Variance	43.11
Min	-19.46
Max	14.11

Near-zero mean: no bias.
 Test SD (6.57) $\approx$ train SE (7.10).

Histogram of Prediction Residual:



Actual vs Predicted Burnout Score



ANOVA: Effect of Categorical Variables on Burnout



One-way ANOVA tested five
categoricals against `burnout_score`:

Variable	F	p-value
<code>attrition_risk</code>	279.2	$< 2e-16$
<code>fear_ai_repl.</code>	1.62	0.197
<code>remote_work</code>	0.94	0.391
<code>ai_adoption</code>	0.47	0.703
<code>company_size</code>	0.21	0.934

Tukey HSD: all three attrition groups
differ at $p < 10^{-6}$

Chi-Square: Fear of AI and Attrition Risk



Are fear and attrition independent?

$$\chi^2 \approx 460, df = 4, p < 2e-98$$

Fear distribution within attrition group:

	Low	Med	High
Low attr.	44%	46%	10%
Med attr.	30%	47%	23%
High attr.	5%	2%	93%

Fear is a *proxy* for attrition risk, not an independent driver of burnout.

Logistic Regression: Predicting High Attrition Risk



Binary outcome: High vs
Low/Medium attrition risk

Model: `glm(family="binomial")`

- burnout_score ↑
- job_satisfaction ↓
- fear_ai (High) ↑↑

Test set accuracy: ≈95%

`step()` retains all three predictors.

Causal chain:

AI usage → burnout → fear →
attrition

Conclusions and Future Paths



Hypothesis confirmed (Step 3):

- All 3 predictors significant at $p < 2e-16$
- $R^2 = 0.611$ – model generalises well
- Job satisfaction is the strongest driver

Categorical analysis (Steps 5–7):

- Only `attrition_risk` significant in ANOVA
- Fear of AI $\leftrightarrow$ attrition: $\chi^2 \approx 460$
- Logistic model: 95% accuracy on High attrition
- `step()` confirms base model is optimal

Future paths:

- 1 Interaction: replacement % $\times$ satisfaction
- 2 Subgroup models by work type / company size
- 3 Longitudinal data for causal inference
- 4 Ordinal logistic for all 3 attrition levels